

Problem Statement

Client M is an e-commerce cosmetics retailer investing across multiple online marketing channels, yet struggling to evaluate the impact of those efforts and optimise their strategy.

Without clear visibility into their **Marketing Return on Investment (MROI)** and each channel’s contribution to sales, budget is being misallocated. Two core problems were identified:

Suboptimal Marketing Strategies

No understanding of which channels drive results or what is working and why.

Inefficient Resource Allocation

Budget dispersed without evidence, preventing optimised spend and reducing overall MROI.

Solution & Objectives

The proposed solution was to build a **Marketing Mix Model (MMM)** using Bayesian Regression, quantifying each channel’s incremental contribution to website users – a reliable proxy for sales.

Primary	Understand & quantify the impact of Marketing Mix elements on Client M’s business performance.
Budget	Optimise budget allocation across channels to maximise ROI.
Acquis.	Identify cost-effective strategies to acquire new customers.
Traffic	Develop recommendations to increase website traffic and MROI.

MMM Workflow CRISP-DM

The project followed the 6-phase CRISP-DM framework iteratively from business scoping through live deployment.

1	Business Understanding	Identify KPIs & marketing goals
2	Data Understanding	Gather & assess historical data
3	Data Preparation	Process & aggregate data sources
4	Modeling	Bayesian Regression MMM
5	Evaluation	Validate model accuracy & fit
6	Deployment	Apply model to optimise spend

Model Input Categories

KPI	Sales volume, market share, CAC, ROI, CLV
Media Data	TV, Digital, Social Media, Print, Radio, Outdoor
Marketing Inputs	Promotions, Email, SEM, Content Marketing, PR, Events
External Factors	Seasonality, economic indicators, competition, industry trends

Key Outcomes

- **Channel ROI Clarity**
Quantified the incremental contribution of every online marketing channel to website users.
- **Budget Reallocation**
Data-driven recommendations surfaced high-ROI channels for increased investment.
- **Customer Acquisition**
Identified the most cost-effective channels for targeted new-user campaigns.
- **Traffic Insights**
Actionable recommendations to grow website traffic and improve overall MROI.

Skills & Tools

Data Analytics	Machine Learning
Marketing Analytics	SQL
Python / IPython	Looker Studio
Power BI	Excel
Data Visualisation	
Bayesian Regression	
Retail & FMCG	English

About This Project

Client	Anonymous (Client M)
Industry	E-Commerce Cosmetics
Model	Bayesian Regression
KPI Proxy	Website Users
Duration	6 Months
Year	2024

Confidentiality

The client’s identity has been anonymised as “Client M” in accordance with a confidentiality agreement. All data, models, and business insights remain proprietary and are not disclosed.